STATISTICS 652

LECTURE 10/10-2006

MULTIVARIATE NORMALITY CHARACTERIZATION

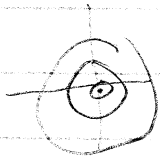
DEF 10.1 If $Z_1, \ldots, Z_q$ ARE MUTUALLY INDEPENDENT $N(0,1)$, then

$\underset{\sim}{Z} = \begin{pmatrix} Z_1 \\ \vdots \\ Z_q \end{pmatrix}$ IS SAID TO HAVE A STANDARD MULTIVARIATE NORMAL DISTRIBUTION

DENOTED: $\underset{\sim}{Z} \sim N_q(\underset{\sim}{0}, I)$

DENSITY: $f(\underset{\sim}{z}) = \prod_{j=1}^{q} (2\pi)^{-\frac{1}{2}} e^{-\frac{1}{2} z_j^2} = (2\pi)^{-\frac{q}{2}} e^{-\frac{1}{2}\sum_{j=1}^{q} z_j^2}$

$= (2\pi)^{-q/2} e^{-\frac{1}{2}\underset{\sim}{z}'\underset{\sim}{z}} \quad \underset{\sim}{z} \in \mathbb{R}^q$



CONCENTRIC SPHERICAL CONTOURS

LEMMA 10.2 If $\underset{\sim}{Z} \sim N_q(\underset{\sim}{0}, I)$, then for $H \in O(q)$, $H\underset{\sim}{Z} \sim N_q(\underset{\sim}{0}, I)$

PROOF: LET $\underset{\sim}{Y} = H\underset{\sim}{Z}$, then

$$f_Y(\underset{\sim}{y}) = f_Z(\underset{\sim}{z}) \Big/ \left|\det\left(\frac{dy_i}{dz_j}\right)\right| = f_Z(H'\underset{\sim}{y}) \Big/ |\det(H)| = f_Z(H'\underset{\sim}{y})$$

$$= (2\pi)^{-q/2} e^{-\frac{1}{2}\underset{\sim}{y}'HH'\underset{\sim}{y}} = (2\pi)^{-\frac{q}{2}} e^{-\frac{1}{2}\underset{\sim}{y}'\underset{\sim}{y}} \quad \underset{\sim}{y} \in \mathbb{R}^q. \quad \square$$

DEF 10.3 $\underset{\sim}{X} \in \mathbb{R}^q$ has a $N_q(\underset{\sim}{\mu}, \Sigma)$ distribution, $\underset{\sim}{\mu} \in \mathbb{R}^q$, $\Sigma_{q\times q} \geq 0$

IFF $X \sim A\underset{\sim}{Z} + \underset{\sim}{\mu}$ where $\underset{\sim}{Z} \sim N_q(\underset{\sim}{0}, I)$ and $\Sigma = AA'$, $A_{q\times q}$.

NOTE: $N_q(\underset{\sim}{\mu}, \Sigma)$ is well defined. THAT IS, distribution depends ONLY ON $\underset{\sim}{\mu}$, $\Sigma = AA'$ and NOT ON THE PARTICULAR SQUARE ROOT A, SINCE

$$A\underset{\sim}{Z} + \underset{\sim}{\mu} \sim AH\underset{\sim}{Z} + \underset{\sim}{\mu} \quad H \in O(q).$$

and $g(A) = g(AH)$ for any $H \in O(q) \Rightarrow g(A) = h(AA')$.

NOTE: FOR $\underset{\sim}{x} \in \mathbb{R}^q$, $g(\underset{\sim}{x}) = g(H\underset{\sim}{x})$ for any $H \in O(q) \Rightarrow g(\underset{\sim}{x}) = h(r)$, $r = \sqrt{\underset{\sim}{x}'\underset{\sim}{x}}$

PROBLEM 3.27

MEAN AND COVARIANCE:

- $\underset{\sim}{Z} \sim N_q(0, I) \Rightarrow E(\underset{\sim}{Z}) = \underset{\sim}{0}$, $VAR(\underset{\sim}{Z}) = I$
- $\underset{\sim}{X} \sim N_q(\underset{\sim}{\mu}, \Sigma) \Rightarrow \underset{\sim}{X} \sim A\underset{\sim}{Z} + \underset{\sim}{\mu} \Rightarrow E(\underset{\sim}{X}) = \underset{\sim}{\mu}$, $VAR(\underset{\sim}{X}) = AA' = \Sigma$
- SO $\underset{\sim}{\mu}$, Σ DENOTE MEAN VECTOR & COVARIANCE MATRIX RESPECTIVELY

DENSITY: $\underset{\sim}{X} \sim N_q(\underset{\sim}{\mu}, \Sigma)$ with $\Sigma > 0$, then

$$f_X(\underset{\sim}{x}) = (2\pi)^{-\frac{q}{2}} |\Sigma|^{-\frac{1}{2}} e^{-\frac{1}{2}(\underset{\sim}{x}-\underset{\sim}{\mu})'\Sigma^{-1}(\underset{\sim}{x}-\underset{\sim}{\mu})} \qquad \underset{\sim}{x} \in \mathbb{R}^p$$

HW: DERIVE VIA 1-1 CHANGE OF VARIABLES ON $Z \sim N_q(0, I)$

- NOTE: CONCENTRIC ELLIPTICAL CONTOURS

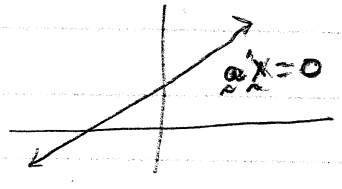


$$f_X(\underset{\sim}{x}) = c \iff (\underset{\sim}{x}-\underset{\sim}{\mu})'\Sigma^{-1}(\underset{\sim}{x}-\underset{\sim}{\mu}) = k^2$$

- NOTE: IF Σ IS SINGULAR, THEN $\underset{\sim}{X}$ DOES NOT HAVE A DENSITY IN $\mathbb{R}^q$. SINGULAR MULTIVARIATE NORMAL

- IN GENERAL, IF $\underset{\sim}{X}$ has a singular covariance matrix, and its distribution function F_X is continuous, then F_X is singular. HOWEVER, THE CONVERSE DOES NOT HOLD.

SINCE, Σ_X SINGULAR $\Rightarrow$ there exist a linear combination $\underset{\sim}{a}'\underset{\sim}{X}$, $\underset{\sim}{a} \neq \underset{\sim}{0}$, such that $VAR(\underset{\sim}{a}'\underset{\sim}{X}) = \underset{\sim}{a}'\Sigma_X\underset{\sim}{a} = 0 \Rightarrow P(\underset{\sim}{a}'\underset{\sim}{X} = c) = 1$ FOR SOME $c \in \mathbb{R}$

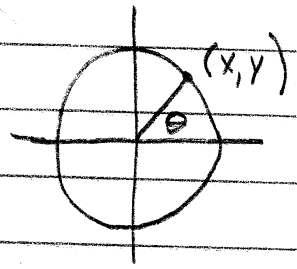


ALL PROBABILITY LIES ON hyperplane $\underset{\sim}{a}'\underset{\sim}{x} = c$

HENCE NO DENSITY IN $\mathbb{R}^2$

COUNTEREXAMPLE TO CONVERSE

$\underset{\sim}{Z} = \begin{pmatrix} X \\ Y \end{pmatrix} \sim$ UNIFORM ON UNIT CIRCLE



i.e. $X = \cos\theta$, $Y = \sin\theta$, $\theta \sim \text{UNIFORM}(0, 2\pi)$

$\Rightarrow \underset{\sim}{Z}$ IS SINGULAR IN $\mathbb{R}^2$ (F_Z CONTINUOUS, BUT NO DENSITY IN $\mathbb{R}^2$)

BY SYMMETRY: $\underset{\sim}{Z} \sim -\underset{\sim}{Z} \Rightarrow \underset{\sim}{\mu}_Z = \underset{\sim}{0}$

ALSO $\begin{pmatrix} X \\ Y \end{pmatrix} \sim \begin{pmatrix} -X \\ Y \end{pmatrix} \Rightarrow \text{cov}(X, Y) = 0$

AND $X \sim Y \Rightarrow \text{VAR}(X) = \text{VAR}(Y) = 1/2$

} VERIFY, WITHOUT USING CALCULUS

NOTE: $X^2 + Y^2 = 1$,

SO $\underset{\sim}{\mu}_Z = \underset{\sim}{0}$, $\underset{\sim}{\Sigma}_Z = \frac{1}{2} I$ which is NON-SINGULAR, ALTHOUGH $\underset{\sim}{Z}$ IS SINGULAR.

THEOREM 10.4 (CLOSURE UNDER LINEAR TRANSFORMATIONS)

$\underset{\sim}{X} \sim N_q(\underset{\sim}{\mu}_X, \underset{\sim}{\Sigma}_X) \Rightarrow \underset{\sim}{Y} = B\underset{\sim}{X} + \underset{\sim}{b} \sim N_p(\underset{\sim}{\mu}_Y, \underset{\sim}{\Sigma}_Y)$ for $B_{p \times q}$, $\underset{\sim}{b} \in \mathbb{R}^p$

where $\underset{\sim}{\mu}_Y = B\underset{\sim}{\mu}_X + \underset{\sim}{b}$ and $\underset{\sim}{\Sigma}_Y = B\underset{\sim}{\Sigma}_X B'$.

PROOF: THE FORM FOR $\underset{\sim}{\mu}_Y$ and $\underset{\sim}{\Sigma}_Y$ hold for any $\underset{\sim}{X} \in \mathbb{R}^q$. TO COMPLETE THE PROOF, WE ONLY NEED TO SHOW $\underset{\sim}{Y}$ IS MULTIVARIATE NORMAL$_p$.

EXPRESS $\underset{\sim}{X} \sim A\underset{\sim}{Z} + \underset{\sim}{\mu}_X$ where $\underset{\sim}{Z} \sim N_p(\underset{\sim}{0}, I)$ AND $\underset{\sim}{\Sigma}_X = AA'$

$\Rightarrow \underset{\sim}{Y} \sim B(A\underset{\sim}{Z} + \underset{\sim}{\mu}_X) + \underset{\sim}{b} = BA\underset{\sim}{Z} + \underset{\sim}{\mu}_Y$ (OR $C\underset{\sim}{Z} + \underset{\sim}{\mu}_Y$ with $C = BA$)

CASE I: $p = q \Rightarrow \underset{\sim}{Y} \sim N_p(\underset{\sim}{\mu}_Y, \underset{\sim}{\Sigma}_Y)$ FOLLOWS FROM DEF. 10.3.

CASE II: $p > q \Rightarrow \underset{\sim}{Y} = (B\ A\ 0)\begin{pmatrix} \underset{\sim}{Z} \\ \underset{\sim}{Z}_0 \end{pmatrix} + \underset{\sim}{\mu}_Y$ where $\begin{pmatrix} \underset{\sim}{Z} \\ \underset{\sim}{Z}_0 \end{pmatrix} \sim N_p(\underset{\sim}{0}, I)$

$\Rightarrow \underset{\sim}{Y} \sim N_p(\underset{\sim}{\mu}_Y, \underset{\sim}{\Sigma}_Y)$ by DEF. 10.3 (NOTE: $\underset{\sim}{\Sigma}_Y$ IS SINGULAR).

CASE III: $p < q$ (e.g. MARGINALS) Let $C = BA = P(\Lambda\ 0)Q'$

DENOTE ITS SINGULAR VALUE DECOMPOSITION

THUS, $\underset{\sim}{Y} \sim BA\underset{\sim}{Z} + \underset{\sim}{\mu}_Y = P(\Lambda\ 0)Q'\underset{\sim}{Z} + \underset{\sim}{\mu}_Y$ Λ is diagonal

$= P(\Lambda\ 0)\underset{\sim}{Z}_0 + \underset{\sim}{\mu}_Y$ where $\underset{\sim}{Z}_0 = Q'\underset{\sim}{Z} \sim N_q(\underset{\sim}{0}, I)$

$= \underbrace{P\Lambda}_{p \times p}\underset{\sim}{Z}_{0,1} + \underset{\sim}{\mu}_Y$ where $\underset{\sim}{Z}_0 = \begin{pmatrix} \underset{\sim}{Z}_{0,1} \\ \underset{\sim}{Z}_{0,2} \end{pmatrix}$ with $\underset{\sim}{Z}_{0,1} \sim N_p(\underset{\sim}{0}, I)$

$\sim N(\underset{\sim}{\mu}_Y, \underset{\sim}{\Sigma}_Y)$ with $\underset{\sim}{\Sigma}_Y = P\Lambda^2P' = CC' = BAA'B'$

$= B\underset{\sim}{\Sigma}_X B'$. □

EMPHASIZE CHARACTERIZATION.

NOTE: THERE WAS NO NEED TO USE CHANGE OF VARIABLES FOR DENSITY TRANSFORMATIONS, WHICH WOULD ONLY APPLY WHEN $\underset{\sim}{\Sigma}_X$ AND $\underset{\sim}{\Sigma}_Y$ ARE NON-SINGULAR.

LEMMA 10.5. (ADDITIVITY OF MULTIVARIATE NORMAL)

If $\underset{\sim}{X}_j \sim N_q(\underset{\sim}{\mu}_j, \underset{\sim}{\Sigma}_j)$ with $\underset{\sim}{X}_1, \ldots, \underset{\sim}{X}_K$ being mutually independent, then $\underset{\sim}{Y} = \underset{\sim}{X}_1 + \cdots + \underset{\sim}{X}_K \sim N_q(\underset{\sim}{\mu}_Y, \underset{\sim}{\Sigma}_Y)$ with

$\underset{\sim}{\mu}_Y = \underset{\sim}{\mu}_1 + \cdots + \underset{\sim}{\mu}_K$ and $\underset{\sim}{\Sigma}_Y = \underset{\sim}{\Sigma}_1 + \cdots + \underset{\sim}{\Sigma}_K$.

PROOF: Express $\underset{\sim}{X}_j \sim A_j \underset{\sim}{Z}_j + \underset{\sim}{\mu}_j$ where $A_j A_j' = \Sigma_j$ and $\underset{\sim}{Z}_1, \ldots, \underset{\sim}{Z}_K$ are mutually independent $N_q(\underset{\sim}{0}, I_q)$. Hence, by Def 10.1,

$\underset{\sim}{Z} = \begin{pmatrix} \underset{\sim}{Z}_1 \\ \vdots \\ \underset{\sim}{Z}_K \end{pmatrix} \sim N_{qK}(\underset{\sim}{0}, I_{qK}) \Rightarrow \underset{\sim}{X} = \begin{pmatrix} \underset{\sim}{X}_1 \\ \vdots \\ \underset{\sim}{X}_K \end{pmatrix}$ is a linear transformation of $\underset{\sim}{Z}$ and hence $\underset{\sim}{X} \sim N_{pK}(\underset{\sim}{\mu}_X, \Sigma_X)$ where $\underset{\sim}{\mu}_X' = (\underset{\sim}{\mu}_1', \ldots, \underset{\sim}{\mu}_K')$ and $\Sigma_X = \begin{bmatrix} \Sigma_1 & 0 & \cdots & 0 \\ 0 & \Sigma_2 & & \\ \vdots & & & \\ 0 & & & \Sigma_K \end{bmatrix}$. The

Lemma follows since $\underset{\sim}{Y} = \underset{\sim}{X}_1 + \cdots + \underset{\sim}{X}_K = B\underset{\sim}{X}$ with $B = \underset{q \times qK}{[I_q \cdots I_q]}$. □

LEMMA 10.6. (Univariate projections)

$\underset{\sim}{X} \sim N_q(\underset{\sim}{\mu}, \Sigma)$ IFF $\underset{\sim}{a}'\underset{\sim}{X} \sim N(\underset{\sim}{a}'\underset{\sim}{\mu}, \underset{\sim}{a}'\Sigma\underset{\sim}{a})$ for all $a \in \mathbb{R}^q$

or for all $\underset{\sim}{a} \in \mathcal{S}^{q-1} = \{\underset{\sim}{a} \in \mathbb{R}^q \mid \underset{\sim}{a}'\underset{\sim}{a} = 1\}$ (q-dim unit sphere).

PROOF: ($\Rightarrow$) By Lemma 10.4 ($\Leftarrow$) by Lemma 7.1 (Cramer-Wold)

MARGINAL DISTRIBUTIONS

$\underset{\sim}{X} = \begin{pmatrix} \underset{\sim}{X}_1 \\ \underset{\sim}{X}_2 \end{pmatrix} \sim N_q(\underset{\sim}{\mu}, \Sigma)$ with $\underset{\sim}{X}_1 \in \mathbb{R}^{q_1}$, $\underset{\sim}{X}_2 \in \mathbb{R}^{q_2}$

$\underset{\sim}{\mu} = \begin{pmatrix} \underset{\sim}{\mu}_1 \\ \underset{\sim}{\mu}_2 \end{pmatrix}$, $\underset{\sim}{\mu}_1 \in \mathbb{R}^{q_1}$, $\underset{\sim}{\mu}_2 \in \mathbb{R}^{q_2}$ and $\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \geq 0$

with Σ_{ij} being $q_i \times q_j$; $i,j = 1,2$.

THEN $\underset{\sim}{X}_1 \sim N_{q_1}(\underset{\sim}{\mu}_1, \Sigma_{11})$ BY LEMMA 10.4 (NO INTEGRATION NEEDED)

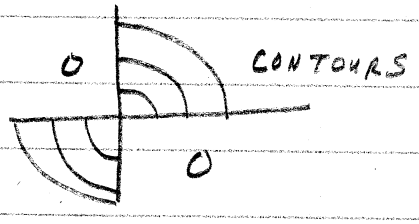
NOTE: NORMAL MARGINALS, EVEN ALL 1-dim, 2-dim, ..., $(q-1)$-dim NORMAL MARGINALS $\not\Rightarrow$ q-dimensional normal.

COUNTER-EXAMPLE: $f(x_1, x_2) = \begin{cases} \frac{1}{\pi} e^{-\frac{1}{2}(x_1^2 + x_2^2)} & \underset{\sim}{x} \in a \\ 0 & \underset{\sim}{x} \notin a \end{cases}$

where $a = \{ \underset{\sim}{x} \in \mathbb{R}^2 \mid x_1 > 0, x_2 > 0 \text{ OR } x_1 < 0, x_2 < 0 \}$

$\Rightarrow X_1 \sim N(0,1)$, $X_2 \sim N(0,1)$

BUT $\underset{\sim}{X}$ IS NOT BIVARIATE NORMAL.



LEMMA 10.7. $\underset{\sim}{X} \sim N_q(\underset{\sim}{\mu}, \Sigma)$ with $\Sigma_{12} = 0$ IFF $\underset{\sim}{X}_1 \sim N_{q_1}(\underset{\sim}{\mu}_1, \Sigma_{11})$, $\underset{\sim}{X}_2 \sim N_{q_2}(\underset{\sim}{\mu}_2, \Sigma_{22})$ and $\underset{\sim}{X}_1, \underset{\sim}{X}_2$ ARE INDEPENDENT.

PROOF FIRST NOTE THAT

$\underset{\sim}{Z} = \begin{pmatrix} \underset{\sim}{Z}_1 \\ \underset{\sim}{Z}_2 \end{pmatrix} \sim N_q(\underset{\sim}{0}, I)$ IFF $\underset{\sim}{Z}_1 \sim N_{q_1}(\underset{\sim}{0}, I)$, $\underset{\sim}{Z}_2 \sim N_{q_2}(\underset{\sim}{0}, I)$ and $\underset{\sim}{Z}_1, \underset{\sim}{Z}_2$ ARE INDEPENDENT. LEMMA FOLLOWS CHARACTERIZING DEF 10.3 AFTER EXPRESING $\underset{\sim}{X} \sim \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \underset{\sim}{Z}$ where

$A_1 A_1' = \Sigma_{11}$ and $A_2 A_2' = \Sigma_{22}$